

Sai Satyabrata

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ABOUT ME

Third-year Computer Science Engineering student with hands-on industry experience at Infosys and DRDO, specializing in cybersecurity, full-stack development, and AI/ML systems. Proficient in Python, Go (Basics), JavaScript, and Rust (Basics), with a strong foundation in system-level programming, distributed systems, and secure software design. Proven track record of delivering scalable, production-grade solutions recognized through a Pre-Placement Offer from DRDO. Actively contributing to open-source communities, including Wikipedia's technical knowledge base. Driven by a passion for building intelligent, high-impact systems at the intersection of software engineering, artificial intelligence, and financial technology.

EXPERIENCE

Python Developer

November 2025 – January 2026

Infosys Springboard

Remote

- Developed and optimized Python-based applications implementing OOP principles, data structures, and algorithms for production-grade software solutions
- Built data analysis pipelines using NumPy, Pandas, and Matplotlib to process and visualize complex datasets, improving decision-making efficiency
- Collaborated with cross-functional teams following Agile methodologies and version control best practices with Git
- Integrated ESG (Environmental, Social, Governance) principles into technology solutions, aligning business objectives with sustainable practices

Cyber Security Analyst

July 2025 – September 2025

The Drop Organization

Remote

- Identified and mitigated 50+ security vulnerabilities across enterprise systems through comprehensive security audits and penetration testing
- Monitored threat intelligence feeds and analyzed suspicious network activities using SIEM tools to prevent potential cyber attacks
- Conducted security compliance checks ensuring adherence to industry standards (ISO 27001, NIST) and organizational security policies
- Developed incident response strategies and risk management frameworks in collaboration with security operations team

EDUCATION

Siksha 'O' Anusandhan University

Bachelor of Technology in Computer Science and Engineering, CGPA: 7.9/10.0

Bhubaneswar, Odisha

September 2024 – August 2027

UCP Engineering School

Diploma in Computer Science and Engineering, CGPA: 7.54/10.0

Berhampur, Odisha

August 2021 – July 2024

Buxi Jagabandhu English Medium School

Matriculation, CGPA: 7.4/10.0

Bhubaneswar, Odisha

August 2021

SKILLS

Programming Languages: Python, JavaScript, SQL, JAVA

Web Technologies: React, Node.js, React Native, HTML/CSS, RESTful APIs

AI/ML Frameworks: TensorFlow, PyTorch, Scikit-learn, Natural Language Processing

Databases & Tools: PostgreSQL, MongoDB, Redis, Git, Linux/Unix

Security Tools: Wireshark, Metasploit, Burp Suite, SIEM, Nmap, Vulnerability Assessment Tools

Core Competencies: Data Structures & Algorithms, System Design, Operating Systems, Computer Networks

Soft Skills: Problem Solving, Team Collaboration, Critical Thinking, Communication, Adaptability, Time Management

PROJECTS

Custom Operating System <i>System-Level Development</i>	Shell-Scripting, Bash-Scripting, x86 Architecture Mar 2024 – Jun 2025
- Architected a custom operating system from scratch with process scheduling, memory management, and file system implementation - Implemented kernel-level components including interrupt handling, system calls, and hardware abstraction layers	
AI Stock Predictor & Trading System Live Website <i>Financial Technology</i>	Python, TensorFlow, JavaScript, CSS Aug 2025 – Jun 2026
- Built an intelligent stock prediction system using LSTM neural networks achieving 85% accuracy on historical market data - Developed algorithmic trading strategies with real-time data processing and automated execution capabilities	
Real-World Airline Pricing Simulator & Booking Platform Live Website <i>Full Stack Web Development</i>	Django, Python Nov 2025 – Jan 2026
- Designed and developed a real-world inspired airline booking system simulating dynamic flight price fluctuations using multiple pricing factors including demand, seat availability, seasonal trends, and departure time - Built advanced backend debugging and monitoring utilities to inspect pricing behavior and validate system integrity, alongside an end-to-end booking flow mimicking enterprise-grade operational tooling	
IRCTC 2.0 Railway Reservation System Live Website <i>Full Stack Web Development</i>	React.js, Next.js, Node.js, MongoDB Feb 2026 – Jun 2026
- Architected a scalable MERN-based reservation platform implementing real-time seat locking with TTL expiration, MongoDB transaction-based booking workflows, and WL → RAC → Confirmed seat promotion to prevent double bookings under concurrent traffic - Engineered secure authentication with JWT and OTP verification, an automatic seat allocation engine, PNR generation, multi-quota reservations, and an admin analytics dashboard with audit logging and rate-limiting middleware	

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

Python Essentials 1 <i>Professional Certification</i>	Cisco Networking Academy Jul 2023
Core Python syntax, data structures, functions, OOP, and file handling	
Cybersecurity Essentials <i>Professional Certification</i>	Cisco Networking Academy Jul 2023
Network security, threat detection, cryptography, and security protocols	
Introduction to Internet of Things (IoT) <i>Professional Certification</i>	Cisco Networking Academy Jul 2023
IoT architecture, sensor integration, edge computing, and connected device protocols	
Artificial Intelligence Job Simulation <i>Industry Simulation Program</i>	Cognizant Dec 2023
ML model development, data preprocessing, and AI-driven business problem solving	
Building Android Apps with React Native <i>Mobile Development Certification</i>	Professional Development Sep 2024
Cross-platform mobile development, component-based UI, and REST API integration	

ACHIEVEMENTS

- Received Pre-Placement Offer (PPO) from DRDO on December 4th, 2025

INTERESTS & HOBBIES

Research & Development: Passionate about R&D in financial markets, algorithmic trading strategies, and quantitative analysis

Retail Trading: Active interest in equity markets, technical analysis, and data-driven investment decision-making

Open Source Contributions: Contributed to the Wikipedia article *Post-quantum cryptography* (Migration section; Blockchain implications section)